

Discrete Kantorovich Formulation

Statements, hints and worked solutions

Each exercise is followed by question-by-question hints and a worked solution. Try the questions before reading the hints.

Exercise 1 (1D discrete $\Rightarrow$ optimal Kantorovich plan).

Let $n = m = 3$, $x_1 < x_2 < x_3$, $y_1 < y_2 < y_3$, and consider two finite measures of mass 3 with weights $a = (1.1, 0.8, 1.1)$ and $b = (1, 1, 1)$. For the quadratic cost $c(x, y) = |x - y|^2$, compute the optimal coupling matrix $P \in \mathbb{R}_+^{3 \times 3}$.

Useful hints

Intersect the source mass intervals $(A_{i-1}, A_i]$ with the target intervals $(B_{j-1}, B_j]$. Their overlap lengths are the entries of the monotone plan.

Detailed solution

In 1D with convex cost, the optimal plan is the monotone (left-to-right) coupling, which one can obtain by the greedy “northwest corner” fill. Cumulative sums are $A = (1.1, 1.9, 3)$ and $B = (1, 2, 3)$. The plan is

$$P = \begin{pmatrix} 1 & 0.1 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0.1 & 1 \end{pmatrix}.$$

Row sums are $(1.1, 0.8, 1.1)$ and column sums $(1, 1, 1)$, as required.

Exercise 2 (Parametric 1D cases).

In both parts use strictly increasing source and target supports and the quadratic cost $|x - y|^2$. The weights below have equal total mass but are not normalized; dividing both marginals and the resulting plan by that mass gives probability measures.

(1) Two against two: $a = (1 + \varepsilon, 1 - \varepsilon)$ and $b = (1, 1)$ with $|\varepsilon| \leq 1$. Compute the optimal $P \in \mathbb{R}_+^{2 \times 2}$ as a function of the sign of ε .

(2) Three against three: $a = (1 + \varepsilon, 1 + \eta, 1 - \varepsilon - \eta)$ and $b = (1, 1, 1)$ with feasibility $1 + \varepsilon \geq 0$, $1 + \eta \geq 0$, $1 - \varepsilon - \eta \geq 0$. Compute the optimal monotone $P \in \mathbb{R}_+^{3 \times 3}$ in the two regimes (i) $\varepsilon > 0$, $\eta > 0$ and (ii) $\varepsilon < 0$, $\eta > 0$.

Useful hints

- (1) Fill the first target mass with the first source, then the second. Include $\varepsilon = 0$.
- (2) Compare the second source cumulative $2 + \varepsilon + \eta$ with 2. This splits regime (ii) into two subcases.

Detailed solution

(1) 2×2 . At $\varepsilon = 0$, $P = I_2$. For nonzero ε , the greedy fill gives two cases.

Case $\varepsilon > 0$. Row 1 first fills column 1 with 1 and sends the excess ε to column 2. Row 2 sends the remaining $1 - \varepsilon$ to column 2:

$$P = \begin{pmatrix} 1 & \varepsilon \\ 0 & 1 - \varepsilon \end{pmatrix}.$$

Case $\varepsilon < 0$. Row 1 has only $1 + \varepsilon$; it fills that much of column 1. Row 2 supplies the remaining $-\varepsilon$ to column 1 and the rest to column 2:

$$P = \begin{pmatrix} 1 + \varepsilon & 0 \\ -\varepsilon & 1 \end{pmatrix}.$$

(2) 3×3 . We proceed left-to-right.

Regime (i): $\varepsilon > 0$, $\eta > 0$ (and thus $\varepsilon + \eta \leq 1$ by feasibility). Row 1 fills column 1 with 1 and sends ε to column 2. Row 2 then completes column 2 with $1 - \varepsilon$ and sends its remainder $\varepsilon + \eta$ to column 3. Row 3 fills the rest of column 3:

$$P = \begin{pmatrix} 1 & \varepsilon & 0 \\ 0 & 1 - \varepsilon & \varepsilon + \eta \\ 0 & 0 & 1 - \varepsilon - \eta \end{pmatrix}.$$

Regime (ii): $\varepsilon < 0$, $\eta > 0$ (feasibility requires $\eta \leq 1 - \varepsilon$). Write $\delta := -\varepsilon > 0$. Row 1 has $1 - \delta$ and sends all of it to column 1. Row 2 supplies the missing δ to column 1, leaving $1 + \eta - \delta = 1 + \eta + \varepsilon$ to place in columns 2–3. There are two subcases depending on whether column 2 can be completed by row 2 alone:

(a) If $\eta \geq \delta$ (i.e. $\eta + \varepsilon \geq 0$): row 2 has at least 1 left for column 2, so it fills column 2 and puts the remaining $\eta - \delta = \eta + \varepsilon$ into column 3; row 3 completes column 3:

$$P = \begin{pmatrix} 1 + \varepsilon & 0 & 0 \\ -\varepsilon & 1 & \eta + \varepsilon \\ 0 & 0 & 1 - \eta - \varepsilon \end{pmatrix}.$$

(b) If $\eta < \delta$ (i.e. $\eta + \varepsilon < 0$): row 2 does not have enough to complete column 2; it contributes $1 + \eta + \varepsilon$ to column 2 and nothing to column 3; row 3 supplies the remaining $\delta - \eta = -(\varepsilon + \eta)$ to column 2 and then fills all of column 3:

$$P = \begin{pmatrix} 1 + \varepsilon & 0 & 0 \\ -\varepsilon & 1 + \eta + \varepsilon & 0 \\ 0 & -(\varepsilon + \eta) & 1 \end{pmatrix}.$$

One checks in each subcase that row sums are $(1 + \varepsilon, 1 + \eta, 1 - \varepsilon - \eta)$ and column sums are $(1, 1, 1)$, with all entries nonnegative under the stated conditions.

Exercise 3 (Couplings in basic cases). (a) What are the couplings between δ_x and δ_y ?

(b) What are the couplings between δ_x and $\frac{1}{2}(\delta_y + \delta_z)$, with $y \neq z$?

(c) Let α, β be finite positive measures with different total masses. What is $\Pi(\alpha, \beta)$, the set of couplings?

Useful hints

(a) Both coordinates are fixed almost surely.

(b) The first coordinate is fixed and the second marginal fixes the two weights.

(c) The total mass of a coupling equals that of each marginal.

Detailed solution

(a) δ_x vs. δ_y . Because $(\text{proj}_1)_\# \pi = \delta_x$, π is supported on $\{x\} \times X$. Because $(\text{proj}_2)_\# \pi = \delta_y$, all mass is at y in the second coordinate. Thus the unique coupling is

$$\pi = \delta_{(x,y)}.$$

(b) δ_x vs. $\frac{1}{2}(\delta_y + \delta_z)$. Any coupling must be supported on $\{x\} \times \{y, z\}$, so

$$\pi = a \delta_{(x,y)} + b \delta_{(x,z)}, \quad a, b \geq 0, \quad a + b = 1.$$

Matching the second marginal $\frac{1}{2}(\delta_y + \delta_z)$ forces $a = b = \frac{1}{2}$. Hence the coupling is unique:

$$\pi = \frac{1}{2} \delta_{(x,y)} + \frac{1}{2} \delta_{(x,z)}.$$

(c) **Different total masses.** Let $M_\alpha := \alpha(X)$ and $M_\beta := \beta(X)$. For any coupling π ,

$$\pi(X \times X) = \alpha(X) = M_\alpha = \beta(X) = \pi(X \times X),$$

so $M_\alpha = M_\beta$ is necessary. If $M_\alpha \neq M_\beta$ then *no* coupling exists:

$$\Pi(\alpha, \beta) = \emptyset.$$

Exercise 4 (Discrete α_t from a coupling).

Let $\alpha_0 = \sum_{i=1}^n a_i \delta_{x_i}$ and $\alpha_1 = \sum_{j=1}^m b_j \delta_{y_j}$ be probability measures on $\mathbb{R}^d$ ($a_i, b_j > 0$, $\sum_i a_i = \sum_j b_j = 1$). Assume the x_i are distinct and the y_j are distinct. Let π be a quadratic-cost optimal coupling between α_0 and α_1 and write it in discrete form

$$\pi = \sum_{i=1}^n \sum_{j=1}^m P_{i,j} \delta_{(x_i, y_j)}, \quad P \in \mathbb{R}_+^{n \times m}, \quad \sum_j P_{i,j} = a_i, \quad \sum_i P_{i,j} = b_j.$$

For $t \in [0, 1]$ define $I_t : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ by $I_t(x, y) = (1-t)x + ty$ and set

$$\alpha_t := (I_t)_\# \pi.$$

- Show that α_t interpolates between α_0 and α_1 , i.e. $\alpha_0 = (P_0)_\# \pi$ and $\alpha_1 = (P_1)_\# \pi$.
- Prove that α_t is discrete for every $t \in [0, 1]$. Give an upper bound on the number of Dirac masses of α_t in terms of P . Deduce the crude bound $\#\text{supp}(\alpha_t) \leq nm$. (Bonus: explain why one can choose an *optimal* coupling with at most $n + m - 1$ nonzeros, hence an interpolation with at most $n + m - 1$ atoms.)
- Give the explicit expression of α_t in terms of the matrix $P \in \mathbb{R}_+^{n \times m}$.
- Assume $n = m$. Under which condition(s) can one choose an optimal coupling so that α_t has exactly n Dirac masses for all t ? Give a concrete example where this happens and write α_t explicitly.

Useful hints

- Evaluate the interpolation map at $t = 0, 1$.
- Push forward each atom separately. For the sharper bound, use an extreme point of the transport polytope.
- Write the image location of each pair (i, j) .
- Equal weights permit a permutation optimizer. Show that two quadratic optimal trajectories cannot collide by using two-cycle monotonicity.

Detailed solution

(a) Endpoints. By definition, $P_0(x, y) = x$ and $P_1(x, y) = y$. Therefore,

$$\alpha_0 = (P_0)_\# \pi = (\text{proj}_X)_\# \pi, \quad \alpha_1 = (P_1)_\# \pi = (\text{proj}_Y)_\# \pi,$$

which equal the marginals of π , hence α_0 and α_1 .

(b) Discreteness and support size. Since π is a finite sum of Dirac masses at points (x_i, y_j) with weights $P_{i,j}$, pushing forward by I_t yields

$$\alpha_t = (I_t)_\# \pi = \sum_{i=1}^n \sum_{j=1}^m P_{i,j} \delta_{(1-t)x_i + ty_j}.$$

Thus α_t is discrete. Each nonzero entry $P_{i,j} > 0$ gives an atom at $(1-t)x_i + ty_j$, so

$$\#\text{supp}(\alpha_t) \leq \#\{(i, j) : P_{i,j} > 0\} \leq nm,$$

with possible strict inequality if different pairs (i, j) produce the same point $(1-t)x_i + ty_j$.

Bonus (sharper bound). The transportation polytope is a nonempty bounded polytope; any basic feasible solution has at most $n + m - 1$ nonzero entries. Hence there exists an optimal coupling $\pi^\star$ with at most $n + m - 1$ nonzeros; using $\pi^\star$ gives an interpolation α_t with at most $n + m - 1$ atoms.

(c) Explicit formula from P . Writing $X \sim \alpha_0, Y \sim \alpha_1$ with joint law π , one has

$$\alpha_t = \mathcal{L}((1-t)X + tY) = \sum_{i=1}^n \sum_{j=1}^m P_{i,j} \delta_{(1-t)x_i + ty_j}.$$

If different pairs $(i, j), (i', j')$ yield the same location, their weights add.

(d) Exactly n atoms. A sufficient condition is a permutation optimizer with $b_{\sigma(i)} = a_i$. In particular, for equal weights $1/n$, Birkhoff's theorem guarantees such an optimizer for any finite cost matrix. For the quadratic cost, its trajectories do not collide.

Indeed, exchanging two matched targets cannot improve the cost, so $\langle x_i - x_k, y_{\sigma(i)} - y_{\sigma(k)} \rangle \geq 0$ for $i \neq k$. If two trajectories collided at $0 < t < 1$, then $y_{\sigma(i)} - y_{\sigma(k)} = -(1-t)(x_i - x_k)/t$, contradicting this inequality since the source points are distinct. At the endpoints distinctness is assumed. Therefore

$$\alpha_t = \sum_i a_i \delta_{(1-t)x_i + ty_{\sigma(i)}}$$

has exactly n atoms for all $t \in [0, 1]$. As a concrete example, for strictly ordered points on the line and uniform weights, take $\sigma(i) = i$; the interpolated locations remain strictly ordered.

Further exercises

Exercise 5 (When does a finite-cost coupling exist?).

Let $n \in \mathbb{N}$ and consider the uniform discrete measures

$$\alpha = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}, \quad \beta = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}.$$

Let $C = (C_{i,j}) \in (\mathbb{R} \cup \{+\infty\})^{n \times n}$ be a cost matrix; $+\infty$ means the pair (x_i, y_j) is forbidden. A coupling π has *finite cost* if it assigns zero mass to all $+\infty$ entries.

(i) 2×2 **by hand**. For

$$C = \begin{pmatrix} C_{1,1} & C_{1,2} \\ C_{2,1} & C_{2,2} \end{pmatrix},$$

determine exactly when a finite-cost coupling exists in terms of the positions of $+\infty$. When it exists, write one explicit coupling matrix $P \in \mathbb{R}_+^{2 \times 2}$ with row/column sums $1/2$ supported only on finite entries.

(ii) **General uniform case**. Let $E = \{(i, j) : C_{i,j} < +\infty\}$ be the set of allowed edges. Prove:

There exists a finite-cost coupling

$$\iff \exists \sigma \text{ a permutation with } (i, \sigma(i)) \in E \text{ for every } i.$$

Useful hints

- (i) There are only two permutation supports.
- (ii) Scale a feasible coupling by n and decompose it into permutation matrices. Nonnegative summands cannot cancel at a forbidden entry.

Detailed solution

(i) 2×2 **case**. There are two permutation supports:

$$\text{Id: } \{(1, 1), (2, 2)\}, \quad \text{Swap: } \{(1, 2), (2, 1)\}.$$

A finite-cost coupling exists iff at least one support is fully finite:

$$(C_{1,1}, C_{2,2} \text{ finite}) \quad \text{or} \quad (C_{1,2}, C_{2,1} \text{ finite}).$$

Obstructions: any all- $+\infty$ row or column; or the pattern blocking both permutations (e.g. exactly one of $\{(1, 1), (2, 2)\}$ and exactly one of $\{(1, 2), (2, 1)\}$ is $+\infty$).

When feasible, an explicit coupling is the scaled permutation on the allowed support. If $(1, 1)$ and $(2, 2)$ are finite, take

$$P = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix};$$

if $(1, 2)$ and $(2, 1)$ are finite, take

$$P = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}.$$

In each case rows and columns sum to $1/2$ and no mass sits on forbidden entries.

(ii) General uniform case.

($\Leftrightarrow$) If there is a permutation σ with $(i, \sigma(i)) \in E$ for all i , then

$$P_{i,\sigma(i)} = \frac{1}{n}, \quad P_{i,j} = 0 \text{ otherwise,}$$

is a coupling with finite cost.

($\Rightarrow$) Suppose a finite-cost coupling exists. Let P be its matrix (row/column sums $1/n$, zeros on forbidden entries) and set $Q := nP$. Then Q is doubly stochastic with $Q_{i,j} = 0$ on forbidden cells. By Birkhoff–von Neumann,

$$Q = \sum_{k=1}^K \lambda_k \Pi_{\sigma_k}, \quad \lambda_k > 0, \quad \sum_k \lambda_k = 1,$$

where Π_{σ_k} are permutation matrices. For any forbidden (i, j) ,

$$0 = Q_{i,j} = \sum_k \lambda_k (\Pi_{\sigma_k})_{i,j},$$

and since each term is nonnegative, we must have $(\Pi_{\sigma_k})_{i,j} = 0$ for all k ; thus every σ_k avoids forbidden entries. As some $\lambda_k > 0$, at least one allowed permutation exists. This proves the equivalence.